



ANDERSON JUNIOR COLLEGE
2009 PRELIMINARY EXAMINATION
HIGHER 2

CHEMISTRY

9746/01

Paper 1 Multiple Choice

18 September 2009

1 hour

Additional Materials : Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid on the answer sheet.

Write your name and PDG.

Write your NRIC / FIN and shade the digits accordingly.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1** Copper, in the form of CuFeS_2 , is extracted from ores in copper mines. The ore contains only 1% by mass of copper.

What percentage of the ore contains CuFeS_2 , if it is the only copper-containing component present?

- A** 1.0 % **B** 2.9 % **C** 3.46 % **D** 34.6 %

- 2** A 25 cm^3 aqueous solution containing Fe^{2+} ions required 36 cm^3 of 0.1 mol dm^{-3} acidified KMnO_4 (aq) for complete reaction.

Calculate the volume of 0.15 mol dm^{-3} acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (aq) that would be needed to react with 10 cm^3 of the same aqueous solution of Fe^{2+} ions.

- A** 8 cm^3 **B** 20 cm^3 **C** 24 cm^3 **D** 40 cm^3

- 3** Gallium nitride, GaN , could revolutionise the design of electric light bulbs because only a small length used as a filament gives excellent light at low cost.

Gallium nitride is an ionic compound containing the Ga^{3+} ion.

What is the electronic configuration of the nitrogen ion in gallium nitride?

- A** $1s^2 2s^2$
B $1s^2 2s^2 2p^3$
C $1s^2 2s^2 2p^4$
D $1s^2 2s^2 2p^6$

- 4** The first eight successive ionization energies, in kJ mol^{-1} , of an unknown element **E** are given below.

420 3100 4400 5900 8000 9600 11400 13300

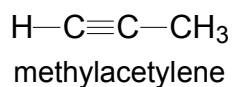
Which group would **E** belong to?

- A** I **B** II **C** VI **D** VII

5 Which of the following species has the correct shape?

- A** H_2S linear
B NCl_3 trigonal planar
C ICl_4^+ square planar
D SbF_5^{2-} square pyramidal

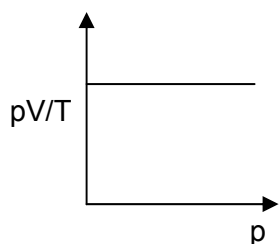
6 What is the total number of σ (sigma) and π (pi) bonds in a methylacetylene molecule?



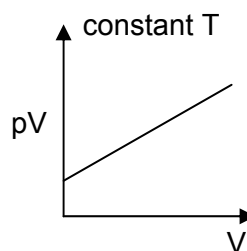
	σ	π
A	4	1
B	5	3
C	6	2
D	7	1

7 Which graph correctly describes the behaviour of a fixed mass of an ideal gas?

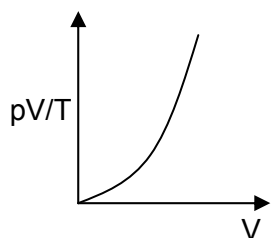
A



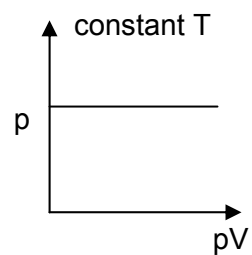
B



C



D



- 8 3 dm³ of helium at a pressure of 5 kPa and 2 dm³ of argon at a pressure of 3 kPa are introduced into a 1.5 dm³ vessel at the same temperature.

What is the total pressure in the vessel?

- A 5 kPa
- B 8 kPa
- C 11.5 kPa
- D 14 kPa

- 9 *The use of the Data Booklet is relevant to this question.*

Given the following enthalpy changes



Which of the following statements is **not** correct?

- A The enthalpy change of atomisation of magnesium is + 167 kJ mol⁻¹.
- B The enthalpy change of formation of Mg²⁺(g) is + 2186 kJ mol⁻¹.
- C The sum of the first and second electron affinities of oxygen is + 649 kJ mol⁻¹.
- D The lattice energy of magnesium oxide is more exothermic than – 3250 kJ mol⁻¹.

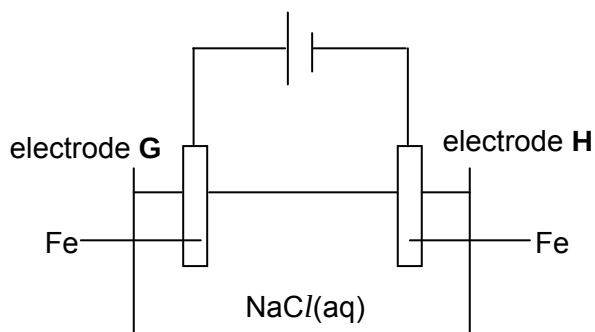
- 10 *The use of the Data Booklet is relevant to this question.*

Which of the following aqueous solution should **not** be stirred with a copper spatula because a reaction could occur?

- A Cr³⁺ (aq)
- B Fe³⁺ (aq)
- C Ni²⁺ (aq)
- D Sn²⁺ (aq)

- 11 The use of the Data Booklet is relevant to this question.

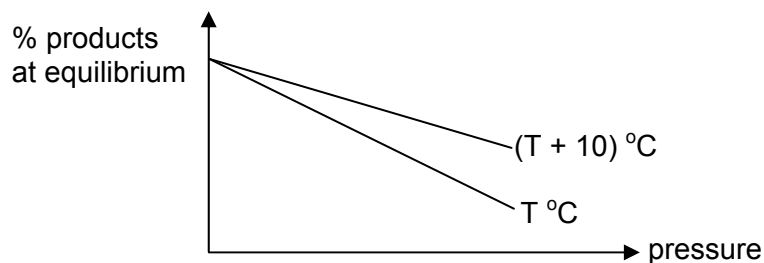
Iron corrodes when it reacts with oxygen to form iron(III) oxide. An experiment was set up as shown below to study the corrosion of iron.



What will occur at electrodes **G** and **H**?

	Electrode G	Electrode H
A	Corrosion of Fe	No corrosion of Fe but Na is deposited
B	Corrosion of Fe	No corrosion of Fe but H ₂ gas is evolved
C	No corrosion of Fe	Corrosion of Fe with O ₂ gas evolved
D	No corrosion of Fe	Corrosion of Fe with H ₂ gas evolved

- 12 The graphs below show how the percentage of gaseous products present at equilibrium vary with temperature and pressure.



Which one of the following reactions could the graph represent?

- A $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g}) \quad \Delta H = + 53 \text{ kJ mol}^{-1}$
- B $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \Delta H = - 92 \text{ kJ mol}^{-1}$
- C $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}) \quad \Delta H = + 57 \text{ kJ mol}^{-1}$
- D $2\text{Fe}(\text{s}) + \frac{3}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{Fe}_2\text{O}_3(\text{s}) \quad \Delta H = - 822 \text{ kJ mol}^{-1}$

- 13** Sodium hydroxide, citric acid and sodium citrate are common ingredients in citrus scented shampoo. The optimum formula of the shampoo has maximum buffering capacity to minimise pH changes to prevent damage to hair.

35 cm³ of 0.2 mol dm⁻³ of sodium hydroxide was mixed with 0.2 mol dm⁻³ of citric acid (monobasic acid).

What is the volume of citric acid required to obtain the optimum formula of the shampoo?

- A** 17.5 cm³
- B** 35 cm³
- C** 42.5 cm³
- D** 70 cm³

- 14** The reaction of nitrogen monoxide and hydrogen gas



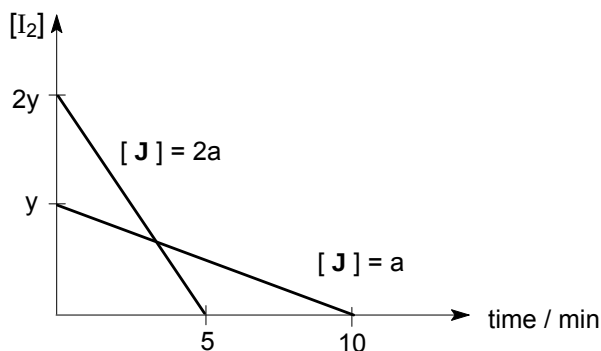
is thought to involve the following steps:

- I** $\text{NO} + \text{NO} \rightleftharpoons \text{N}_2\text{O}_2$ (fast)
- II** $\text{N}_2\text{O}_2 + \text{H}_2 \longrightarrow \text{H}_2\text{O} + \text{N}_2\text{O}$ (slow)
- III** $\text{N}_2\text{O} + \text{H}_2 \longrightarrow \text{N}_2 + \text{H}_2\text{O}$ (fast)

Which of the following is true about the reaction?

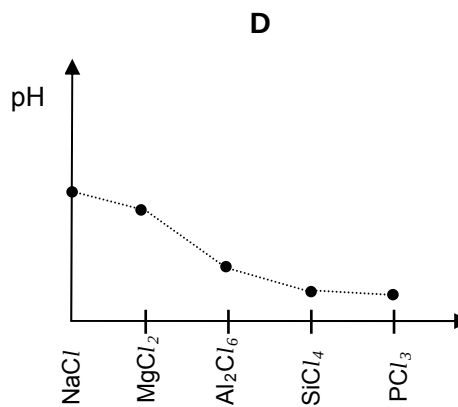
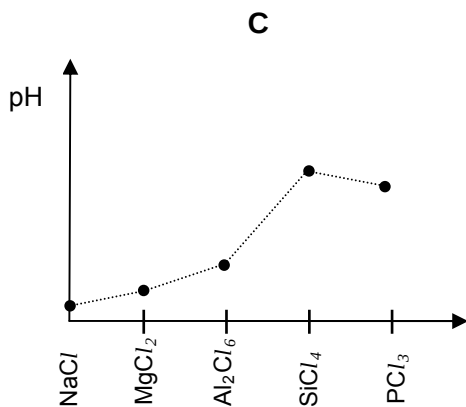
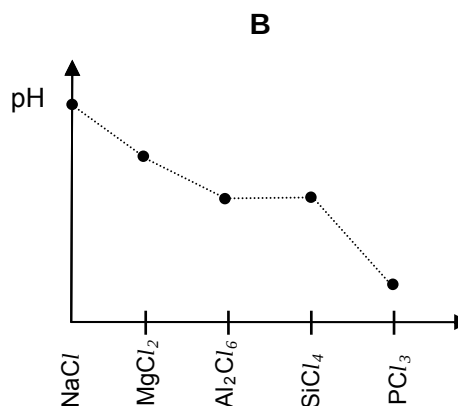
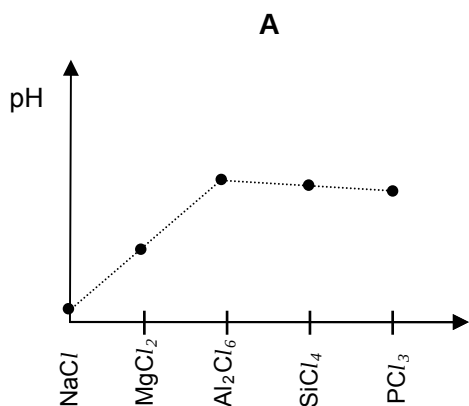
- A** H₂ acts as the catalyst.
- B** The rate equation for the reaction is rate = $k [\text{N}_2\text{O}_2] [\text{H}_2]$.
- C** Increasing the concentration of NO does not change the rate of reaction.
- D** The overall order of the reaction is 3.

- 15 The kinetics of the reaction between iodine and compound **J** is investigated.



What conclusions can be drawn from the graphs?

- A The reaction is second order with respect to compound **J** because rate of reaction increases by four times when its concentration is increased by two times.
- B Both iodine and compound **J** are involved in the rate determining step.
- C The reaction is first order with respect to iodine because half-life is constant.
- D The overall order of the reaction is 1.
- 16 The chlorides of the elements sodium to phosphorous are separately added to water. Which of the following diagrams best represents the pH of the solutions produced?

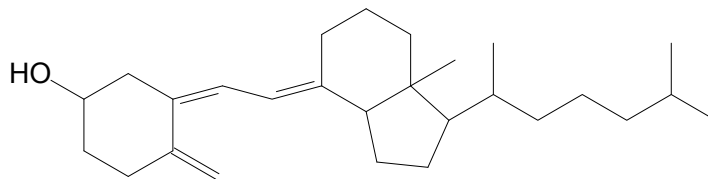


- 17 Which one of the following statements about beryllium, the element above magnesium in Group II, is true?
- A Beryllium reacts rapidly with cold water.
- B Beryllium oxide is soluble in water.
- C Beryllium nitrate decomposes at a higher temperature than magnesium nitrate.
- D Beryllium shows a fixed oxidation state of +2 in its compounds.
- 18 Astatine, At, is a halogen below iodine in Group VII of the Periodic Table. Which statement is most likely to be true for astatine?
- A Hydrogen astatide is stable to heat.
- B Copious fumes of hydrogen astatide are observed when hot concentrated sulfuric acid reacts with potassium astatide.
- C Silver astatide is insoluble in dilute aqueous ammonia.
- D Iodine is produced when aqueous potassium iodide is added to astatine.
- 19 A compound of chromium with the general formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ forms an aqueous solution. When 0.1 mol of this compound is treated with excess silver nitrate, 0.1 mol of silver chloride is formed.

Which of the following represents the structure of the chromium ion present in the compound?

- A $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- B $[\text{CrCl}(\text{H}_2\text{O})_5]^{2+}$
- C $[\text{CrCl}_2(\text{H}_2\text{O})_4]^+$
- D $[\text{CrCl}_3(\text{H}_2\text{O})_3]$

- 20 The diagram shows the structure of *cholecalciferol*, which is a form of vitamin D.



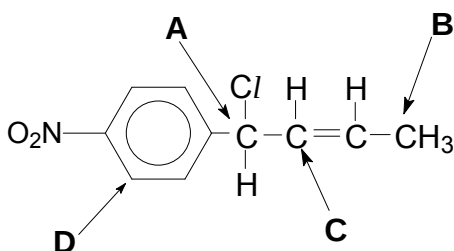
How many chiral carbon atoms are present in this molecule?

- A 4 B 5 C 6 D 7

- 21 Which of the following gives the largest number of isomers on reacting with hot ethanolic potassium hydroxide?

- A $(\text{CH}_3)_3\text{CCH}_2\text{Cl}$
 B $\text{CH}_3\text{CH}_2\text{CHClCH}_3$
 C $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)\text{CH}_2\text{Cl}$
 D $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)\text{Cl}$

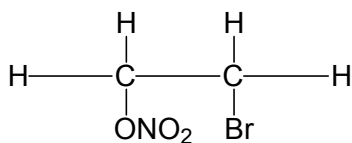
- 22 In the molecule shown below, which carbon atom is most likely to be attacked by a nucleophile?



- 23 Straight chain alkanes undergo cracking at high temperatures and in the presence of catalyst. Which of the following does **not** represent a correct cracking reaction?

- A $\text{C}_{10}\text{H}_{22} \rightarrow \text{C}_3\text{H}_8 + \text{C}_5\text{H}_{10} + \text{C}_2\text{H}_4$
 B $\text{C}_{13}\text{H}_{28} \rightarrow \text{C}_4\text{H}_8 + \text{C}_6\text{H}_{14} + \text{CH}_3\text{CH}=\text{CH}_2$
 C $\text{C}_7\text{H}_{16} \rightarrow \text{CH}_3\text{CH}=\text{CH}_2 + \text{CH}(\text{CH}_3)_3$
 D $\text{C}_{11}\text{H}_{24} \rightarrow \text{C}_4\text{H}_{10} + \text{CH}_2=\text{CH}_2 + \text{C}_5\text{H}_{10}$

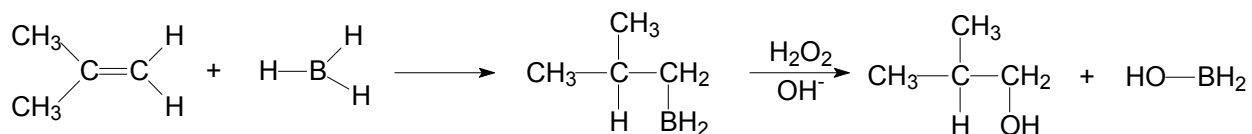
- 24 The following compound is produced when ethene reacts with bromine in the presence of concentrated aqueous sodium nitrate.



Which of the following descriptions about the organic intermediate in this reaction is correct?

- A It has a negative charge.
 B It is a free radical.
 C It contains carbon, hydrogen and bromine atoms.
 D It contains carbon, hydrogen, oxygen and nitrogen atoms.

- 25 Boranes, such as BH_3 , are readily added across a carbon-carbon double bond and the resulting product can be oxidised with alkaline H_2O_2 .

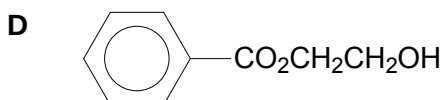
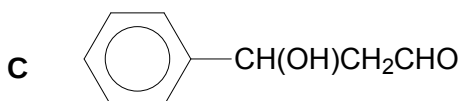
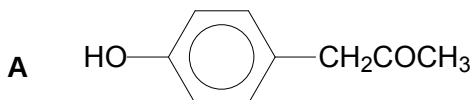


Which compound is produced when 3-methylpent-2-ene is treated with BH_3 , followed by alkaline H_2O_2 ?

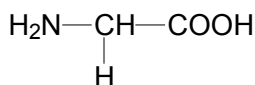
- A $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)\text{OH}$
 B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{CH}_3)_2$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2\text{OH}$
 D $(\text{CH}_3\text{CH}_2)_2\text{C}(\text{CH}_3)\text{OH}$
- 26 Which of the following salts would produce the most alkaline solution when dissolved in water?
- A $\text{C}_6\text{H}_5\text{O}^-\text{Na}^+$
 B $\text{C}_2\text{H}_5\text{O}^-\text{Na}^+$
 C $\text{C}_2\text{H}_5\text{CO}_2^-\text{Na}^+$
 D $\text{C}_2\text{H}_5\text{NH}_3^+\text{Cl}^-$
- 27 Which procedure gives the best yield of phenylbenzoate, $\text{C}_6\text{H}_5\text{CO}_2\text{C}_6\text{H}_5$?
- A Reacting $\text{C}_6\text{H}_5\text{CO}_2\text{H}$ with SOCl_2 , then adding $\text{C}_6\text{H}_5\text{OH}$.
 B Reacting $\text{C}_6\text{H}_5\text{OH}$ with NaOH , then adding $\text{C}_6\text{H}_5\text{COCl}$.
 C Refluxing a mixture $\text{C}_6\text{H}_5\text{CO}_2\text{H}$ and $\text{C}_6\text{H}_5\text{OH}$ with concentrated H_2SO_4 .
 D Reacting $\text{C}_6\text{H}_5\text{OH}$ with SOCl_2 , then adding $\text{C}_6\text{H}_5\text{CO}_2\text{H}$.

- 28 When an organic compound **L** was treated with phosphorous pentachloride, fumes of hydrogen chloride were evolved. When **L** was warmed with Fehling's solution, a brick-red precipitate was obtained.

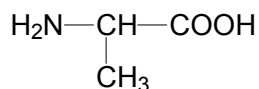
Which of the following is likely to be compound **L**?



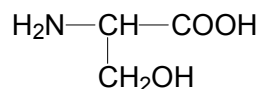
- 29 Silk fibroin is a type of fibrous protein made up almost entirely of β -pleated sheets. The main amino acids found in silk fibroin are alanine, glycine, serine and cysteine. The structures of these amino acids are given below.



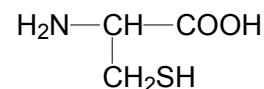
glycine



alanine



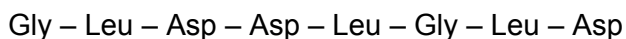
serine



cysteine

Which of the following statements about the type of interactions present in silk fibroin is **not** correct?

- A The β -pleated sheets are held together by hydrogen bonding.
 B There are van der Waals' forces between glycine and alanine in the protein.
 C There are disulphide linkages present in silk fibroin.
 D Serine forms ionic bonds with another serine amino acid in the protein.
- 30 How many different tripeptides can be produced from the partial hydrolysis of the following octapeptide (eight amino acid residues)?



A 3

B 4

C 5

D 6

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of these statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which statements about a 28.0 g sample of $^{14}\text{N}_2$ are correct?

- 1** The number of molecules is half the number of atoms in 32.0 g of $^{16}\text{O}_2$.
- 2** The number of atoms is twice the number of atoms in 4.0 g of ^4He .
- 3** The number of atoms is the same as the number of atoms in 24.0 g of ^{12}C .

32 The Gas Laws can be summarised in the ideal gas equation $pV = nRT$, where each symbol has its usual meaning.

Which of the following statements are correct?

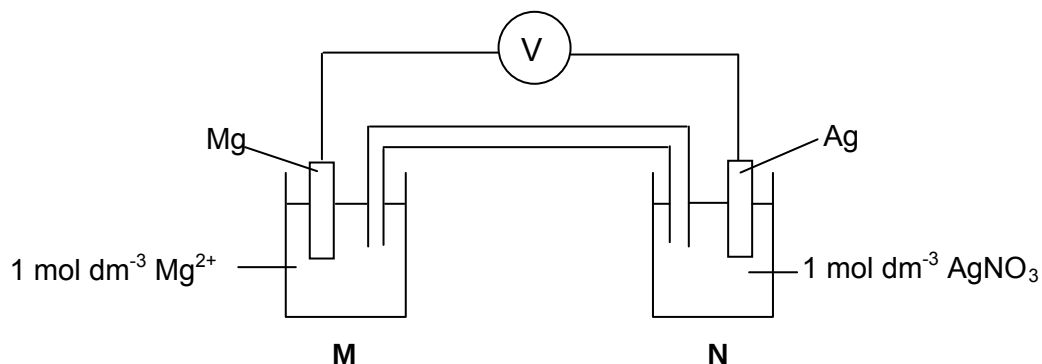
- 1** One mole of any ideal gas occupies the same volume under the same conditions of temperature and pressure.
- 2** The density of an ideal gas at constant pressure is inversely proportional to the temperature.
- 3** The volume of a given mass of an ideal gas is doubled if its temperature is raised from 25 °C to 50 °C at constant pressure.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

33 Which of the following statements are true for the electrochemical cell below?



- 1** The electrons flow from half-cell **M** to **N**.
- 2** The $E^\ominus_{\text{cell}}$ value decreases when a small amount of NaCl (aq) is added to the half-cell **N**.
- 3** The $E^\ominus_{\text{cell}}$ value increases when a small amount of KOH (aq) is added to the half-cell **M**.

34 Which of the following statements about the Period 3 elements are true?

- 1** Atomic radii decrease from sodium to chlorine.
- 2** Their oxides become more acidic from sodium to chlorine.
- 3** The maximum oxidation state is shown by silicon.

35 Magnesium sulfate is more soluble than barium sulfate because

- 1** the ionic radius of Ba^{2+} is larger than that of Mg^{2+} .
- 2** the enthalpy change of hydration of Mg^{2+} is more exothermic than that of Ba^{2+} .
- 3** the ionic radius of the sulfate ion is smaller than that of the cations, Mg^{2+} and Ba^{2+} .

The responses **A** to **D** should be selected on the basis of

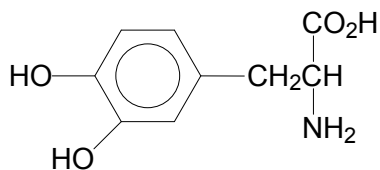
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

36 Which of the following can act as a ligand?

- 1** AlH_4^-
- 2** NH_2NH_2
- 3** CH_3OH

37 *Dopamine* is a neurotransmitter found in many animals, including vertebrates and invertebrates. Which of the following statements are correct?



dopamine

- 1** 1 mole of *dopamine* reacts with 3 moles of acyl chloride
- 2** 1 mole of *dopamine* reacts with 1 mole of HBr .
- 3** 1 mole of *dopamine* reacts with 3 moles of Na_2CO_3

38 Which of the following reactions yields a carbon compound incorporating deuterium, D? [D = ^2H]

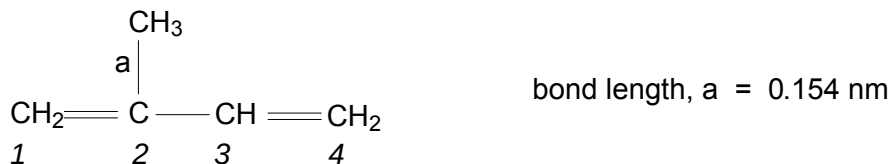
- 1** $\text{C}_6\text{H}_5\text{COBr} \xrightarrow{\text{D}_2\text{O}}$
- 2** $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{LiAlD}_4}$
- 3** $\text{CH}_3\text{NHCOCH}_3 \xrightarrow{\text{NaOD, D}_2\text{O}}$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

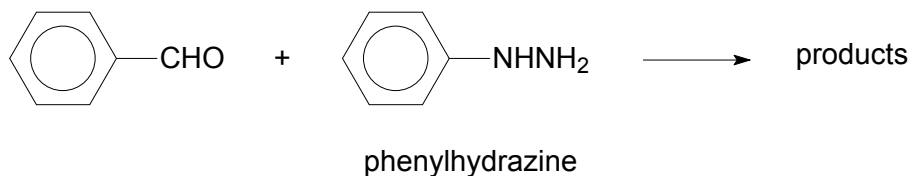
- 39** 2-methylbuta-1,3-diene can be polymerised to make synthetic rubbers.



From the structure shown, which statements about 2-methylbuta-1,3-diene are correct?

- The bond between carbons 2 and 3 is formed by the overlap of two sp^2 orbitals.
 - The bond length between carbons 2 and 3 is 0.154 nm.
 - When treated with aqueous acidified $\text{K}_2\text{Cr}_2\text{O}_7$, a compound with four alcohol groups is formed.
- 40** The reaction of phenylhydrazine is similar to that of 2,4-dinitrophenylhydrazine.

Which changes in bonding occur during the reaction below?



	Bond broken	Bond formed
1	C–H	C–C
2	N–H	O–H
3	C=O	C=N